The Compatibility Equations

Statement and Proof

1 Setting and notation

Let $\Omega \subset \mathbb{R}^3$ be an open, connected domain and $\boldsymbol{u} : \Omega \to \mathbb{R}^3$ a displacement field of class C^3. Under the small-strain hypothesis, the linearised strain tensor $\underset{\sim}{\varepsilon}$ is

$$\varepsilon_{ij} \;=\; \tfrac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) = \tfrac{1}{2}\left(u_{i,j} + u_{j,i}\right), \tag{1}$$

where a comma denotes partial differentiation and repeated indices are summed from 1 to 3. $\underset{\sim}{\varepsilon}$ is symmetric: $\varepsilon_{ij} = \varepsilon_{ji}$.

Equation (1) gives six strain components in terms of only three displacement components, so a prescribed symmetric field $\underset{\sim}{\varepsilon}$ does not, in general, derive from any displacement. The *compatibility equations* are the conditions $\underset{\sim}{\varepsilon}$ must satisfy for such a $\boldsymbol{u}$ to exist.

2 Statement

Theorem 1. *Let $\underset{\sim}{\varepsilon}$ be a symmetric tensor field of class C^2 on Ω, and consider the equations*

$$\boxed{\frac{\partial^2 \varepsilon_{ij}}{\partial x_k\,\partial x_k} + \frac{\partial^2 \varepsilon_{kk}}{\partial x_i\,\partial x_j} - \frac{\partial^2 \varepsilon_{jk}}{\partial x_i\,\partial x_k} - \frac{\partial^2 \varepsilon_{ik}}{\partial x_j\,\partial x_k} = 0, \qquad \textit{equivalently} \qquad \operatorname{Inc}(\underset{\sim}{\varepsilon}) = \operatorname{rot}\operatorname{rot}(\underset{\sim}{\varepsilon}) = 0.} \tag{2}$$

(a) ***Necessity.*** *If $\underset{\sim}{\varepsilon}$ derives from a displacement $\boldsymbol{u} \in C^3(\Omega)$ through (1), then (2) holds.*

(b) ***Sufficiency.*** *If Ω is simply connected and $\underset{\sim}{\varepsilon}$ satisfies (2), there exists $\boldsymbol{u} \in C^3(\Omega)$ satisfying (1), unique up to an infinitesimal rigid-body displacement $\boldsymbol{a} + \boldsymbol{b} \times \boldsymbol{x}$.*

Throughout, we use the following auxiliary quantities:

$$R_{ij} := \varepsilon_{ij,kk} + \varepsilon_{kk,ij} - \varepsilon_{jk,ik} - \varepsilon_{ik,jk} \qquad \text{(left-hand side of (2))}, \tag{3}$$

$$\eta_{mn} := e_{mik}\, e_{njl}\, \varepsilon_{ij,kl} = \left(\operatorname{rot}\operatorname{rot}\underset{\sim}{\varepsilon}\right)_{mn} \qquad \text{(incompatibility tensor)}, \tag{4}$$

$$C_{ijkl} := \varepsilon_{ij,kl} + \varepsilon_{kl,ij} - \varepsilon_{ik,jl} - \varepsilon_{jl,ik} \qquad \text{(Saint-Venant tensor)}, \tag{5}$$

where e_{mik} is the Levi-Civita permutation symbol. In (4), rot is applied once on each index of $\underset{\sim}{\varepsilon}$, so $\operatorname{rot}\operatorname{rot}\underset{\sim}{\varepsilon}$ is a symmetric second-order tensor. Note that $R_{ij} = C_{ijkk}$ (set $l = k$ in (5) and sum).

3 Proof of necessity

Proof of (a). Since $\boldsymbol{u} \in C^3$, partial derivatives commute (Schwarz), so the order of indices after the comma is irrelevant. From (1):

$$\varepsilon_{ij,kk} = \tfrac{1}{2}\left(u_{i,jkk} + u_{j,ikk}\right), \qquad \varepsilon_{kk,ij} = u_{k,kij},$$
$$\varepsilon_{jk,ik} = \tfrac{1}{2}\left(u_{j,ikk} + u_{k,ijk}\right), \qquad \varepsilon_{ik,jk} = \tfrac{1}{2}\left(u_{i,jkk} + u_{k,ijk}\right).$$

Hence

$$R_{ij} = \tfrac{1}{2}u_{i,jkk} + \tfrac{1}{2}u_{j,ikk} + u_{k,ijk} - \tfrac{1}{2}u_{j,ikk} - \tfrac{1}{2}u_{k,ijk} - \tfrac{1}{2}u_{i,jkk} - \tfrac{1}{2}u_{k,ijk} = 0.$$

The same computation applied to (5) gives $2C_{ijkl} = (u_{i,jkl} - u_{i,kjl}) + (u_{j,ikl} - u_{j,lik}) + (u_{k,lij} - u_{k,ijl}) + (u_{l,kij} - u_{l,jik}) = 0$, so the full Saint-Venant equations $C_{ijkl} = 0$ hold as well; by Section 4, so does $\operatorname{Inc}(\underset{\sim}{\varepsilon}) = 0$. □

4 Equivalence of the three forms

Proposition 1. *For any symmetric C^2 field $\underset{\sim}{\varepsilon}$ in dimension 3,*

$$R = 0 \iff \operatorname{Inc}(\underset{\sim}{\varepsilon}) = 0 \iff C = 0.$$

Proof. (i) $R \leftrightarrow \eta$. Expand the product of permutation symbols as a determinant of Kronecker deltas:

$$e_{mik}e_{njl} = \delta_{mn}(\delta_{ij}\delta_{kl} - \delta_{il}\delta_{kj}) - \delta_{mj}(\delta_{in}\delta_{kl} - \delta_{il}\delta_{kn}) + \delta_{ml}(\delta_{in}\delta_{kj} - \delta_{ij}\delta_{kn}).$$

Contracting with $\varepsilon_{ij,kl}$ and using the symmetry of $\underset{\sim}{\varepsilon}$:

$$\eta_{mn} = \delta_{mn}\big(\varepsilon_{ii,kk} - \varepsilon_{ik,ik}\big) - \varepsilon_{mn,kk} + \varepsilon_{im,in} + \varepsilon_{nk,km} - \varepsilon_{ii,mn} = \delta_{mn}\big(\Delta \operatorname{tr} \underset{\sim}{\varepsilon} - \operatorname{div}\operatorname{div} \underset{\sim}{\varepsilon}\big) - R_{mn}.$$

Taking the trace of (3) gives $\operatorname{tr} R = 2\big(\Delta \operatorname{tr} \underset{\sim}{\varepsilon} - \operatorname{div}\operatorname{div} \underset{\sim}{\varepsilon}\big)$, so

$$\eta = -R + \tfrac{1}{2}(\operatorname{tr} R)\, I, \qquad \text{and, taking the trace,} \qquad R = -\eta + (\operatorname{tr}\eta)\, I. \tag{6}$$

Therefore $R = 0 \iff \eta = 0$.

(ii) $\eta \leftrightarrow C$. Using $\varepsilon_{ij} = \varepsilon_{ji}$ and commutation of derivatives, one checks directly from (5):

$$C_{ijkl} = -C_{kjil} = -C_{ilkj} = C_{klij},$$

i.e. C is antisymmetric in (i,k) and in (j,l). Contracting (5) with $e_{mik}e_{njl}$, the last two terms (symmetric in (i,k)) drop out and the first two give equal contributions, so

$$e_{mik}e_{njl}\, C_{ijkl} = 2\eta_{mn}.$$

Conversely, an object antisymmetric in (i,k) and (j,l) is recovered from this contraction via $e_{mik}e_{mab} = \delta_{ia}\delta_{kb} - \delta_{ib}\delta_{ka}$:

$$\tfrac{1}{2}\, e_{mik}e_{njl}\, \eta_{mn} = \tfrac{1}{4}(\delta_{ia}\delta_{kb} - \delta_{ib}\delta_{ka})(\delta_{jc}\delta_{ld} - \delta_{jd}\delta_{lc})\, C_{acbd} = \tfrac{1}{4}\big(C_{ijkl} - C_{ilkj} - C_{kjil} + C_{klij}\big) = C_{ijkl}.$$

Thus $C = 0 \iff \eta = 0$. □

Remark 1. In 3D, C has 81 components but, by its symmetries, only 6 are independent: those of the symmetric tensor η. This is why (2), which has 6 components (R is symmetric), already contains the full Saint-Venant system.

5 The six equations written out

Writing (x, y, z) for (x_1, x_2, x_3), the diagonal components of $\mathrm{Inc}(\underset{\sim}{\varepsilon})$ give

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2} = 2\,\frac{\partial^2 \varepsilon_{xy}}{\partial x\,\partial y}, \qquad \frac{\partial^2 \varepsilon_{yy}}{\partial z^2} + \frac{\partial^2 \varepsilon_{zz}}{\partial y^2} = 2\,\frac{\partial^2 \varepsilon_{yz}}{\partial y\,\partial z}, \qquad \frac{\partial^2 \varepsilon_{zz}}{\partial x^2} + \frac{\partial^2 \varepsilon_{xx}}{\partial z^2} = 2\,\frac{\partial^2 \varepsilon_{zx}}{\partial z\,\partial x}, \tag{7}$$

and the off-diagonal components give

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y\,\partial z} = \frac{\partial}{\partial x}\left(-\frac{\partial \varepsilon_{yz}}{\partial x} + \frac{\partial \varepsilon_{zx}}{\partial y} + \frac{\partial \varepsilon_{xy}}{\partial z}\right), \tag{8}$$

$$\frac{\partial^2 \varepsilon_{yy}}{\partial z\,\partial x} = \frac{\partial}{\partial y}\left(-\frac{\partial \varepsilon_{zx}}{\partial y} + \frac{\partial \varepsilon_{xy}}{\partial z} + \frac{\partial \varepsilon_{yz}}{\partial x}\right), \tag{9}$$

$$\frac{\partial^2 \varepsilon_{zz}}{\partial x\,\partial y} = \frac{\partial}{\partial z}\left(-\frac{\partial \varepsilon_{xy}}{\partial z} + \frac{\partial \varepsilon_{yz}}{\partial x} + \frac{\partial \varepsilon_{zx}}{\partial y}\right). \tag{10}$$

(For instance $\eta_{zz} = e_{zik}e_{zjl}\varepsilon_{ij,kl} = \varepsilon_{xx,yy} + \varepsilon_{yy,xx} - 2\varepsilon_{xy,xy}$.) In plane strain only the first one remains: $\partial^2\varepsilon_{xx}/\partial y^2 + \partial^2\varepsilon_{yy}/\partial x^2 = 2\,\partial^2\varepsilon_{xy}/\partial x\,\partial y$.

6 Proof of sufficiency

Lemma 1 (Poincaré)**.** *Let Ω be simply connected and $f_1, f_2, f_3 \in C^1(\Omega)$ with $f_{k,l} = f_{l,k}$. Then there exists $\varphi \in C^2(\Omega)$ with $\varphi_{,k} = f_k$, unique up to an additive constant, given by $\varphi(\boldsymbol{x}) = \varphi(\boldsymbol{x}_0) + \int_{\boldsymbol{x}_0}^{\boldsymbol{x}} f_k\,dx_k$ (path-independent).*

Proof of (b). By Proposition 1, the hypothesis (2) implies $C_{ijkl} = 0$ for all indices. We reconstruct first the rotation, then the displacement.

Motivation. If $\boldsymbol{u}$ exists, write $u_{i,j} = \varepsilon_{ij} + \omega_{ij}$ with $\omega_{ij} = \frac{1}{2}(u_{i,j} - u_{j,i})$ skew-symmetric. Then

$$\omega_{ij,k} = \tfrac{1}{2}\,(u_{i,jk} - u_{j,ik}) = \varepsilon_{ik,j} - \varepsilon_{jk,i}, \tag{11}$$

since $\varepsilon_{ik,j} - \varepsilon_{jk,i} = \frac{1}{2}(u_{i,kj} + u_{k,ij} - u_{j,ki} - u_{k,ji})$. The gradient of ω depends on $\underset{\sim}{\varepsilon}$ only; we reverse this.

Step 1: the rotation ω. For fixed (i, j) let $f_k^{(ij)} := \varepsilon_{ik,j} - \varepsilon_{jk,i}$. Then

$$f_{k,l}^{(ij)} - f_{l,k}^{(ij)} = \varepsilon_{ik,jl} + \varepsilon_{jl,ik} - \varepsilon_{il,jk} - \varepsilon_{jk,il} = C_{ikjl} - C_{iljk},$$

as one sees by expanding both C's with (5) (the terms $\varepsilon_{ij,kl}$ and $\varepsilon_{kl,ij}$ cancel). This vanishes, so by Lemma 1 there exist $\omega_{ij} \in C^2(\Omega)$ with $\omega_{ij,k} = \varepsilon_{ik,j} - \varepsilon_{jk,i}$ and $\omega_{ij}(\boldsymbol{x}_0) = \omega^0_{ij}$, where ω^0 is any skew matrix. Since $f^{(ji)} = -f^{(ij)}$ and ω^0 is skew, uniqueness gives $\omega_{ji} = -\omega_{ij}$.

Step 2: the displacement $\boldsymbol{u}$. For fixed i let $g_j^{(i)} := \varepsilon_{ij} + \omega_{ij}$. Then

$$g_{j,k}^{(i)} = \varepsilon_{ij,k} + \varepsilon_{ik,j} - \varepsilon_{jk,i},$$

which is symmetric in (j, k). By Lemma 1 there exists $u_i \in C^3(\Omega)$ with $u_{i,j} = \varepsilon_{ij} + \omega_{ij}$ and $u_i(\boldsymbol{x}_0) = a_i$ arbitrary.

Step 3: verification. By skew-symmetry of ω: $\frac{1}{2}(u_{i,j} + u_{j,i}) = \frac{1}{2}(\varepsilon_{ij} + \omega_{ij} + \varepsilon_{ji} + \omega_{ji}) = \varepsilon_{ij}$.

Step 4: uniqueness. If $\boldsymbol{u}, \boldsymbol{u}'$ both give $\underset{\sim}{\varepsilon}$, then $\boldsymbol{w} = \boldsymbol{u} - \boldsymbol{u}'$ has zero strain; by (11) its rotation has zero gradient, hence is a constant skew matrix on the connected set Ω, and $w_{i,j} = \omega_{ij}$ gives $w_i = a_i + \omega_{ij}x_j$, i.e. $\boldsymbol{w} = \boldsymbol{a} + \boldsymbol{b} \times \boldsymbol{x}$. These are the six free constants ω^0, $\boldsymbol{a}$ of Steps 1–2. □

Remark 2 (Role of simple connectivity). On a multiply connected domain (e.g. a hollow cylinder) the line integrals of Lemma 1 may depend on the path around a hole: a field satisfying (2) may then give a *multivalued* displacement (Volterra dislocations). Single-valuedness requires in addition that $\oint f_k^{(ij)}\,dx_k = 0$ and $\oint g_j^{(i)}\,dx_j = 0$ around each independent loop.